

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Jnana Sangama, Belagavi - 590 018



PROJECT PHASE II DIARY ON

“PrintChakra – AI-Powered Smart Print and Capture Solution”

Submitted in partial fulfilment of the requirements for the VII Semester (BAI786)

Bachelor of Engineering

in

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Submitted By

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JANATHA EDUCATION SOCIETY

VIVEKANANDA INSTITUTE OF TECHNOLOGY

Gudimavu, Kumbalgodu(P), Kengeri(H), Bengaluru -560074

2025-2026

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND.00. MACHINE LEARNING

PrintChakra – AI-Powered Smart Print and Capture Solution

Project Diary

Team Members:

Chaman S – 1VK22AI007

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Team Guide:

Dr. Shaila K,

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VI Semester: Major Project Phase I

Sl No.	Date	Work Summary	Hours Worked	Outcome	Guide's Sign
1	15/03/2025	Identified problems in traditional print-scan workflows such as manual scanning, scanner dependency, and internet reliance. Finalized problem statement and objectives.	4	Clear understanding of real-world document digitization challenges.	
2	22/03/2025	Conducted literature survey on document digitization systems and OCR-based automation techniques.	5	Gained theoretical knowledge of OCR-based systems.	
3	29/03/2025	Studied OpenCV fundamentals including image preprocessing, edge detection, and noise reduction.	6	Built foundation in computer vision techniques.	
4	05/04/2025	Analyzed perspective correction and document alignment techniques for camera-based scanning.	5	Improved understanding of document enhancement methods.	
5	12/04/2025	Studied OCR engines with focus on Tesseract OCR and multilingual text extraction.	5	Understood OCR workflow and text recognition challenges.	
6	19/04/2025	Designed overall system architecture including mobile capture, desktop backend, database, and dashboard modules.	6	Learned modular system architecture design.	
7	26/04/2025	Prepared block diagrams and data-flow diagrams for offline-first workflow.	5	Improved system visualization and	

Sl No.	Date	Work Summary	Hours Worked	Outcome	Guide's Sign
				planning skills.	
8	05/05/2025	Developed basic Flask backend structure and REST API endpoints.	6	Hands-on experience with backend framework setup.	
9	15/05/2025	Configured MongoDB with GridFS for storing scanned images, OCR text, and metadata.	6	Learned large-file storage and metadata handling.	
10	25/05/2025	Implemented initial OpenCV-based document scanning pipeline.	5.5	Practical exposure to image processing implementation.	
11	05/06/2025	Integrated Tesseract OCR with backend and tested text extraction accuracy.	6	Gained experience in OCR integration.	
12	15/06/2025	Implemented local Wi-Fi based communication between mobile app and desktop backend.	5.5	Understood offline data transfer mechanisms.	

VII Semester: Major Project Phase II

Sl No.	Date	Work Summary	Hours Worked	Outcome	Guide's Sign
13	05/07/2025	Implemented offline synchronization with retry logic, acknowledgements, and local queuing.	6	Improved reliability of offline-first synchronization.	
14	15/07/2025	Enhanced image preprocessing pipeline for better OCR accuracy.	5.5	Improved text recognition performance.	
15	05/08/2025	Developed React.js dashboard for document listing and management.	6	Improved frontend development skills.	
16	15/08/2025	Added document preview, OCR text viewing, and metadata display features in dashboard.	5.5	Enhanced dashboard usability.	
17	05/09/2025	Implemented document export features such as PDF and text download.	5	Learned file handling and export logic.	
18	15/09/2025	Implemented JWT-based authentication and role-based access control.	6	Strengthened system security.	

Sl No.	Date	Work Summary	Hours Worked	Outcome	Guide's Sign
19	25/09/2025	Configured SSL-secured communication between modules.	5	Improved secure data transmission understanding.	
20	10/10/2025	Implemented detailed logging for system activities and document operations.	5.5	Learned importance of audit trails.	
21	25/10/2025	Conducted extensive OCR accuracy and synchronization testing.	6.5	Identified performance issues and edge cases.	
22	10/11/2025	Optimized API response times and image processing workflow.	5.5	Improved system efficiency.	
23	20/11/2025	Performed end-to-end system validation across mobile and desktop components.	6	Verified stable system operation.	
24	05/12/2025	Prepared final documentation including diagrams, screenshots, and results.	4.5	Improved technical documentation skills.	
25	15/12/2025	Completed final review, future scope analysis, and submission preparation.	4	Ensured project completion and readiness for evaluation.	

Signature of the HOD

Dr. Shaila K

Prof. & HOD

Department of AIML

Signature of the Principal

Dr. K M Ravi Kumar

Principal

Bengaluru VKIT, Bengaluru